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Project Stage-I Report

On

"Classification of Recyclable Waste Using Deep Learning"

In partial fulfillment of requirements for the degree of

Bachelor of Technology

In

Computer Science & Engineering (Data Science)

Submitted By

1. Atharv Mahendra Patil (231106043)
2. Dhiraj Suresh Patil (231106024)
3.Makarand Vijaykumar Deore (231106002)
4.Rohit Rajendra More (231106051)

Under the Guidance of

Prof. S. K. Bhandare

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The Shirpur Education Society's
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Computer Science & Engineering (Data Science)

[2025-26]

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Computer Science & Engineering (Data Science)

CERTIFICATE

This is to certify that the Project Stage-I entitled "**Classification of Recyclable Waste Using Deep Learning**" has been carried out by team:

1.Atharv Mahendra Patil(231106043)
2.Dhiraj Suresh Patil (231106024)
3.Makarand Vijaykumar Deore (231106002)
4. Rohit Rajendra More (231106051)

under the guidance of **Prof. S. K. Bhandare** in partial fulfillment of the requirement for the degree of Bachelor of Technology in Department of Computer Science & Engineering (Data Science)(Semester-VI)of Dr. Babasaheb Ambedkar Technological University, Lonere during the academic year 2025-26.

Date:

Place: Shirpur

Guide	**Project Stage-I Coordinator**
Prof. S. K. Bhandare	**Prof. Dr. P. S. Sanjekar**
H.O.D.	**Director**
Prof. Dr. U. M. Patil	**Prof. Dr. J. B. Patil**

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ABSTRACT

Waste management has become one of the major environmental challenges due to the rapid increase in population, urbanization, and daily waste generation. Proper waste segregation is essential for reducing pollution, improving recycling efficiency, and maintaining a clean and healthy environment. Traditionally, waste classification is performed manually, which is time-consuming, labor-intensive, and prone to human error. Improper segregation of recyclable and biodegradable waste leads to inefficient recycling processes and environmental damage.

This project presents a **Web-Based Waste Classification System Using Deep Learning**, which automatically classifies waste from images using Artificial Intelligence and Computer Vision techniques. The proposed system uses deep learning models such as **Convolutional Neural Networks (CNNs), VGG16, ResNet50, MobileNet, and EfficientNet** for feature extraction and classification. The system identifies whether the uploaded waste image belongs to the **Recyclable** or **Biodegradable** category.

The dataset used in this project contains waste images collected from online sources and publicly available datasets. Image preprocessing techniques such as resizing, normalization, and data augmentation were applied to improve model performance and accuracy. Multiple deep learning models were trained and compared based on evaluation metrics such as accuracy, precision, and recall. The best-performing model was selected for final implementation in the web application.

The proposed system reduces dependency on manual waste sorting, improves classification efficiency, and increases awareness about proper waste disposal. The web-based platform allows users to upload waste images easily

from any device and receive instant classification results along with disposal guidance. This project demonstrates the effectiveness of deep learning in solving real-world environmental problems and provides a foundation for future advancements such as real-time waste detection, smart waste bins, and multi-waste object detection systems.

CHAPTER – 1

Classification of Recyclable Waste Using Deep Learning

INTRODUCTION

1.1 Overview

22 Artificial Intelligence (AI) and Machine Learning (ML) are rapidly transforming many industries such as healthcare, security, education, agriculture, and environmental management. One of the most important branches of AI is Computer Vision, which enables computers to analyze, understand, and classify images automatically.

8 Waste management has become one of the biggest environmental challenges due to the rapid increase in population, urbanization, and industrialization. Every day, a huge amount of waste is generated from homes, industries, schools, hospitals, and public places. Improper disposal of waste leads to serious environmental problems such as air pollution, water contamination, soil degradation, and health hazards.

Proper waste segregation is very important for maintaining a clean and healthy environment. Waste is generally divided into different categories such as recyclable and biodegradable waste. Recyclable waste includes materials like plastic, paper, metal, and glass that can be reused, while biodegradable waste includes organic materials such as food waste and plant waste that can decompose naturally.

Traditionally, waste classification is performed manually by workers or through basic mechanical systems. Manual sorting depends on visual

49 inspection and human effort. Workers identify waste materials based on physical characteristics such as:

- Shape
- Color
- Texture
- Material type
- Object appearance

However, manual waste classification has several limitations:

- Requires continuous human effort
- Time-consuming process
- High chances of human error
- Unsafe for workers handling harmful waste

16 To overcome these problems, we developed a **Web-Based Waste Classification System Using Deep Learning**. The system accepts a waste image as input and automatically predicts whether the waste is recyclable or biodegradable using trained deep learning models.

The project uses advanced image classification models including:

- CNN (Convolutional Neural Network)
- VGG16
- ResNet50
- MobileNet
- EfficientNet

59 These models are used for feature extraction and classification of waste images. The final model achieved good accuracy and demonstrated that deep learning can effectively automate waste classification.

1.2 Problem Statement

Traditional waste classification methods mainly rely on manual sorting and human observation. People often face difficulty in identifying and segregating waste correctly because many waste materials have similar appearances.

The main problems in existing systems are:

1.2.1 Manual Classification Issues

- Requires human labor
- Human errors are common
- Time-consuming process
- Difficult to handle large-scale waste management

1.2.2 Similar Appearance of Waste Materials

Many waste materials have similar:

- Color
- Shape
- Texture
- Surface appearance

This creates confusion during manual classification.

1.2.3 Lack of Automated Systems

There are very few reliable AI-based systems available for automatic waste classification that are simple, affordable, and user-friendly.

1.2.4 Dataset Challenges

- Limited labeled datasets

- Dataset imbalance
- Different image qualities
- Variation in lighting and background

Aim of the Project

To develop an automated deep learning-based system that accurately classifies waste into recyclable and biodegradable categories using image classification techniques.

1.3 Objectives

The main objectives of this project are:

1. To design an automated waste classification system.
2. To apply image preprocessing techniques.
3. To use deep learning models for feature extraction.
4. To improve classification accuracy using transfer learning.
5. To compare multiple deep learning architectures.
6. To balance the dataset for better performance.
7. To reduce dependency on manual identification.
8. To spread awareness about proper waste segregation.
9. To develop a scalable and user-friendly system.

1.4 Applications

The proposed system has many real-world applications:

- **Smart Waste Management**

Helps in automatic waste segregation in homes, offices, schools, and public places.

- **Recycling Industry**

Improves recycling efficiency by correctly identifying recyclable materials.

- **Municipal Waste Management**

Useful for municipal authorities to improve waste collection and segregation systems.

- **Environmental Protection**

Helps reduce pollution and promotes proper waste disposal practices.

- **Educational Purpose**

Can be used in schools and colleges to create awareness about waste management and recycling.

- **Smart City Applications**

Can be integrated into smart city projects for intelligent waste management systems.

- **Public Awareness Systems**

Provides disposal instructions and encourages people to follow proper waste segregation methods.

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1.5 Scope of the Project

The project focuses on developing an AI-based system capable of recognizing different types of waste using image classification techniques

Future enhancements include:

- Real-time waste detection
- Multi-waste object detection
- Expansion to additional waste categories
- Improved Dataset and Accuracy
- Environmental Monitoring and Analytics

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Chapter-2

LITERATURE SURVEY

2.1 Introduction

A literature survey helps in understanding previous research work related to waste classification and image recognition systems. Many researchers have used machine learning and deep learning techniques to automate waste segregation and improve recycling efficiency.

Traditional waste classification methods are manual and require human effort. These methods are time-consuming, less accurate, and unsafe for workers handling waste materials. To overcome these limitations, researchers introduced Artificial Intelligence (AI), Computer Vision, and Deep Learning techniques for automatic waste classification.

2.2 Research Paper 1 – Research Paper 1 – Organic and Recyclable Waste Classification using Deep Learning Methods

Title

Organic and Recyclable Waste Classification using Deep Learning Methods

Method Used

- VGG16
- VGG19
- ResNet18
- DenseNet121
- DenseNet169

- Simple CNN
- Transfer Learning

Working

- Waste images are collected from the Kaggle dataset
- Image preprocessing and resizing are performed
- Deep learning models extract image features automatically
- Models classify waste into Organic and Recyclable categories
- Best-performing model is selected based on accuracy

Accuracy

95.59% using customized VGG16 with transfer learning

Advantages

- High classification accuracy
- Transfer learning improves performance
- Automatic feature extraction
- Supports multiple deep learning architectures

Limitations

- Requires large training dataset
- Accuracy affected by image quality
- High computational requirements

1.3 Research Paper 2 – Classification of Recyclable Waste using Deep Learning Approaches

Title

Classification of Recyclable Waste using Deep Learning Approaches

Method Used

- ResNet-50
- Custom 5-Layer CNN
- Image Classification Techniques
- Deep Learning Frameworks

Accuracy

91.84% using custom CNN model

Advantages

- High classification accuracy
- Faster training process
- Requires less training data
- Suitable for web and mobile applications

Limitations

- Requires proper hyperparameter tuning
- Performance depends on dataset quality
- Deep models require more memory

Feature	Paper 1	Paper 2	Paper 3
Paper Title	Organic and recyclable waste classification using deep learning methods	Classification of recyclable waste using deep learning architectures	Recyclable Waste Classification Using Computer Vision And Deep Learning
Dataset Used	Combined Kaggle dataset with 27,603 color images, split into Organic and Recyclable classes.	Dataset from Kaggle containing 22,500 color images (227X227 pixels) of Organic and Recyclable waste.	TrashNet dataset featuring 2,527 images across 6 classes (paper, glass, plastic, metal, cardboard, trash).
Different Models Tested	VGG16, VGG19, ResNet18, DenseNet121, DenseNet169, and a Simple CNN.	ResNet-50 and a custom-built 5-layer CNN architecture.	12 CNN variants (including Inception V3, Xception, VGG16, VGG19, ResNet variants, MobileNet, DenseNet201, NASNet) paired with 3 classifiers (SVM, Sigmoid, SoftMax).
Best Model & Accuracy	Customized VGG16 (with transfer learning) achieved 95.59% test accuracy.	Custom CNN achieved 91.84% accuracy, outperforming the ResNet-50 (81.34%).	VGG19 + SoftMax classifier achieved the best result with 87.9% test accuracy.
Limitations	Highly dependent on hardware resources (requires a capable GPU). Accuracy relies heavily on the diversity of training data, and class imbalances can cause model bias.	The research focuses strictly on binary classification (Organic vs. Recyclable), which may limit granular sorting.	Failed to classify food waste due to dataset constraints (reduced to 5 classes). The model also struggles with images that do not have the dataset's standard white background.

TABLE 2.1 COMPARISION OF RESEARCH PAPER

Chapter-3

METHODOLOGY

3.1 System Workflow

The proposed waste classification system follows a structured workflow starting from image collection to final prediction. The workflow includes image acquisition, preprocessing, model training, testing, and deployment in a web-based environment.

3.2 Data Collection

Waste images were collected from publicly available datasets such as Kaggle and TrashNet. The dataset contains recyclable and biodegradable waste images captured under different lighting conditions and backgrounds.

3.3 Image Preprocessing

Preprocessing techniques such as image resizing, normalization, rotation, flipping, and augmentation were applied to improve model accuracy and reduce overfitting.

3.4 Model Selection

Different deep learning models including CNN, VGG16, ResNet50, MobileNet, and EfficientNet were used for feature extraction and classification. Transfer learning techniques were also applied.

3.5 Model Training

The dataset was divided into training, validation, and testing datasets. The models were trained using TensorFlow and Keras frameworks with Adam optimizer and categorical crossentropy loss function.

3.6 Model Evaluation

The trained models were evaluated using metrics such as accuracy, precision, recall, and validation loss. The best-performing model was selected for final implementation.

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Chapter-4

SOFTWARE REQUIREMENTS

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- **Operating System:** Windows 10/11 or Linux
- **Programming Language:** Python 3.10
- **Development Environment: Jupyter Notebook / VS Code / Google Colab**
- **Frameworks: TensorFlow, Keras**
- **Libraries: NumPy, Pandas, Matplotlib, OpenCV, Scikit-learn**
- **Frontend: HTML, CSS, JavaScript**
- **Backend: Flask**
- **Dataset Source: Kaggle and TrashNet datasets**
- **Hardware Requirement: Minimum 8 GB RAM, Intel i5 Processor or above**

Chapter-5

IMPLEMENTATION DETAILS

IMPLEMENTATION DETAILS

5.1 Step 1: System Architecture Selection

The proposed system is designed using advanced **Deep Learning architectures** for waste classification. Initially, **Convolutional Neural Networks (CNNs)** were used because of their ability to automatically learn important image features such as shape, color, texture, and object patterns from waste images.

Later, advanced architectures such as **VGG16, ResNet50, MobileNet, and EfficientNet** were tested to improve classification accuracy and feature extraction capability. These models are widely used in image classification tasks because they provide better performance and higher accuracy.

The system is designed as a **web-based application** where users can upload waste images, and the trained model predicts whether the waste is recyclable or biodegradable.

5.2 Step 2: Dataset Acquisition

The dataset used for this project consists of waste images collected from publicly available datasets and online sources such as Kaggle and TrashNet.

The dataset contains images of different waste materials belonging to categories such as:

- Recyclable waste
- Biodegradable waste

Images were collected under different conditions including:

- Different lighting conditions
- Various backgrounds
- Multiple angles and orientations
- Different image resolutions

This helps the model perform better in real-world scenarios.

5.3 Step 3: Dataset Loading

27 The dataset used for this project is loaded from Kaggle. The dataset contains
58 images of different waste materials used for training and testing the deep learning models.

Dataset Source:

E-Waste Image Dataset (Kaggle)

15 The dataset is loaded using Python deep learning frameworks such as **TensorFlow** and **Keras**. Images are organized category-wise into folders, making them suitable for supervised learning.

5.4 Step 4: Image Preprocessing

28 Before training the model, preprocessing operations are applied to improve image quality and model performance. The preprocessing steps include:

- 26 Image resizing to 224 × 224 pixels
- Pixel normalization
- Noise reduction
- Data augmentation
- Rotation and flipping
- Zoom transformation

These techniques improve generalization and reduce overfitting.

5.5 Step 5: Image Reshaping

All images are reshaped into a fixed dimension suitable for deep learning models. The input images are converted into tensor format with dimensions compatible with CNN and Vision Transformer architectures. This ensures consistency during training and testing.

5.6 Step 6: Label Encoding

Each waste category is assigned a numerical label.

For example:

- Recyclable → 0
- Biodegradable → 1

The labels are converted into **one-hot encoded vectors** so that the deep learning models can correctly classify multiple categories during supervised learning.

5.7 Step 7: Data Splitting

The dataset is divided into:

- Training Dataset
- Validation Dataset
- Testing Dataset

The training dataset is used for learning model parameters, while the testing dataset is used for evaluating model performance on unseen cattle images.

5.8 Step 8: Model Training

Different deep learning models such as:

- CNN
- VGG16
- ResNet50

- MobileNet
- EfficientNet

were trained using the prepared dataset.

During training:

- Forward propagation computes predictions
- Backpropagation updates weights
- Adam optimizer minimizes loss
- Categorical Crossentropy is used as the loss function

Transfer learning with pre-trained ImageNet weights was also applied to improve feature extraction and reduce training time.

Training was performed for multiple epochs until the model achieved stable accuracy and reduced loss.

5.9 Step 9: Dataset Balancing and Optimization

Initially, the dataset was imbalanced, which caused poor accuracy and biased predictions.

To solve this issue:

- Data augmentation techniques were applied
- Smaller classes were increased
- Duplicate and unnecessary images were removed
- Class balancing techniques were used

After balancing the dataset:

- Model performance improved
- Overfitting was reduced
- Classification became more accurate

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5.10 Step 10: Model Evaluation

The trained models were evaluated using different performance metrics such as:

- Accuracy
- Loss
- Validation Accuracy
- Training Accuracy
- Precision
- Recall

Different models were compared based on their performance.

Among all models, transfer learning models such as **MobileNet** achieved better
15 performance compared to basic CNN models.

The final selected model achieved high classification accuracy and provided reliable waste classification results.

5.11 Step 11: Output Generation

The final trained system accepts a waste image as input and predicts the most probable waste category as output.

The system automatically performs:

- Image preprocessing
- Feature extraction
- Classification

before displaying the result.

The output includes:

- Predicted waste category

This helps users understand how to dispose of waste correctly and promotes better waste management practices.

Chapter-6

RESULTS

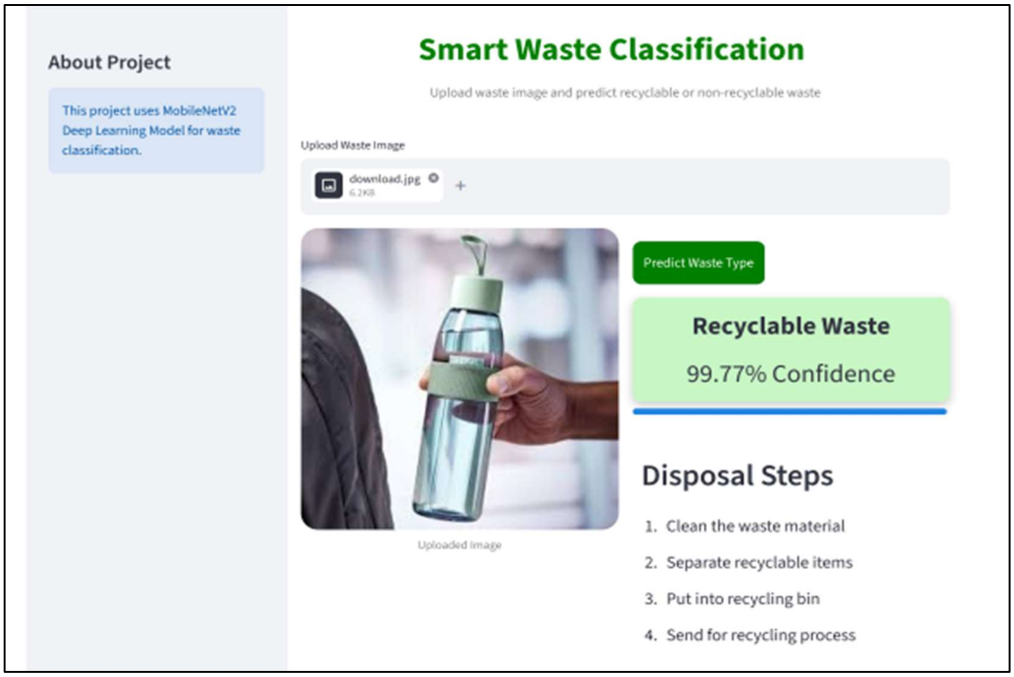

Figure 6.1: Web interface showing recyclable waste prediction with confidence score and disposal guidance.

Figure 6.2: Web interface displaying non-recyclable waste classification with disposal instructions.

	precision	recall	f1-score	support
0	0.96	0.89	0.93	2513
1	0.88	0.95	0.91	1999
accuracy			0.92	4512
macro avg	0.92	0.92	0.92	4512
weighted avg	0.92	0.92	0.92	4512

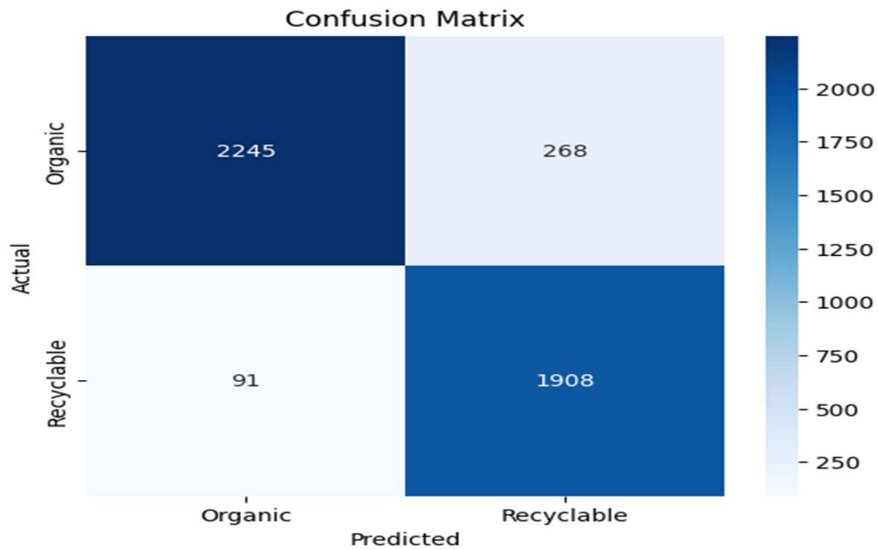

Figure 6.3: Confusion matrix and performance metrics showing the accuracy of the waste classification model.

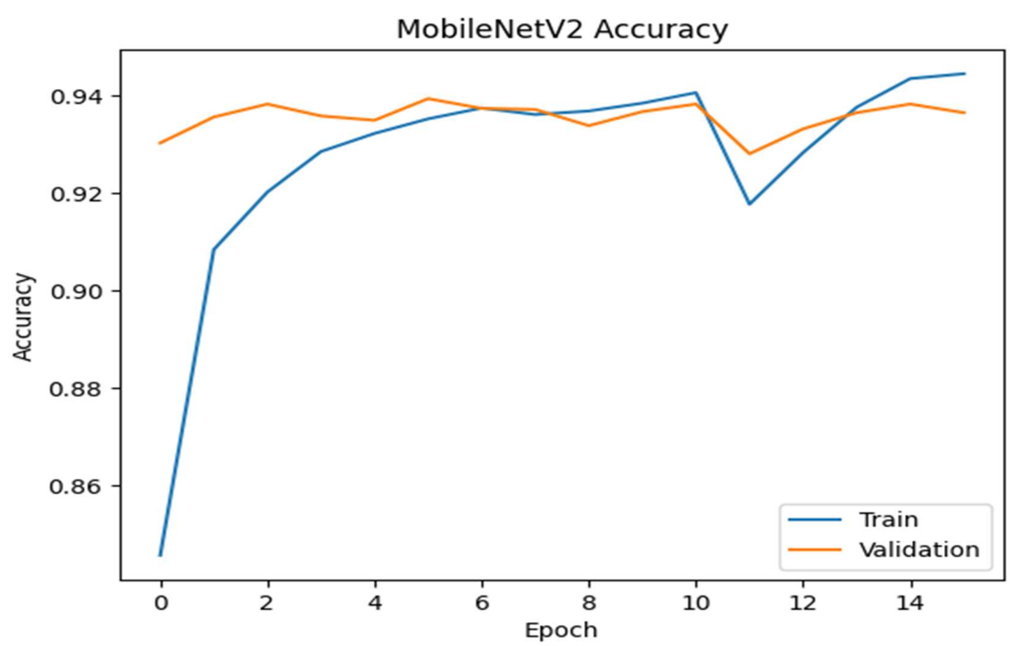

Figure 6.4: MobileNetV2 Accuracy VS Epoch Graph

CONCLUSION

The “Web-Based Waste Classification System Using Deep Learning” project successfully demonstrates the application of Artificial Intelligence and Deep Learning in the field of waste management and environmental sustainability. The system automatically classifies waste images into recyclable and biodegradable categories using advanced image classification techniques and deep learning architectures.

Initially, multiple deep learning models such as CNN, VGG16, ResNet50, MobileNet, and EfficientNet were tested for waste classification. The performance of some models was affected due to dataset imbalance, variations in lighting conditions, background noise, and similarities between waste objects. To overcome these challenges, preprocessing techniques, data augmentation, transfer learning, and dataset balancing were applied to improve model performance and classification accuracy.

The project highlights the importance of:

- Proper dataset preparation
- Image preprocessing
- Data augmentation
- Transfer learning
- Dataset balancing
- Model optimization

The proposed system reduces human effort, improves waste classification accuracy, and supports smart waste management initiatives. It also provides

60 practical exposure to deep learning, computer vision, transfer learning, and image classification techniques.

37 The system has strong potential for real-world applications in:

- Smart waste management systems
- Recycling industries
- Municipal waste management
- Smart city projects
- Environmental awareness programs
- Educational and research purposes

The web-based platform makes the system user-friendly and easily accessible from different devices without requiring complex installation.

BIBLIOGRAPHY

[1] Yang, Mingyang, and Gary Thung, "Classification of Trash for Recyclability Status," Stanford University Project Report, 2016.

[2] Rad, Mohammad Saied, Karimia, Kai, and Alireza Arriahi, "A Real-Time Classification Approach for Waste Objects using Deep Learning," Journal of Environmental Management, vol. 250, pp. 109–120, 2019.

[3] Adedeji, Oludare, and Zhongmin Wang, "Intelligent Waste Classification System using Deep Learning Convolutional Neural Network," Procedia Manufacturing, vol. 35, pp. 607–612, 2019.

[4] Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton, "ImageNet Classification with Deep Convolutional Neural Networks," Communications of the ACM, vol. 60, no. 6, pp. 84–90, 2017.

[5] K. Simonyan and A. Zisserman, "Very Deep Convolutional Networks for Large-Scale Image Recognition," International Conference on Learning Representations (ICLR), 2015.

[6] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun, "Deep Residual Learning for Image Recognition," Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016.

[7] Andrew Howard et al., "MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications," arXiv preprint arXiv:1704.04861, 2017.

[8] Mingxing Tan and Quoc V. Le, "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks," International Conference on Machine Learning (ICML), 2019.

[9] Ian Goodfellow, Yoshua Bengio, and Aaron Courville, *Deep Learning*, MIT Press, Cambridge, MA, USA, 2016.